

Uber Supply Demand Gap Analysis Report

Prepared By: Ankita Mukund Panpatil

Tools Used: Microsoft Excel, MySQL, Python (Pandas, NumPy, Matplotlib)

1. Project Objective

The objective of this project is to analyze Uber ride request data and identify the supply-demand gap by examining trip status, pickup locations, request times, and ride completion patterns.

2. Dataset Information

Dataset: Uber Request Data

Total Records: 6745

Original Columns: 6

Additional Columns: Hour, Day, Time Slot

3. Excel Analysis

- Data Cleaning
- Missing Value Check
- Duplicate Check
- Hour, Day and Time Slot creation
- Pivot Tables
- Charts
- Interactive Dashboard

4. SQL Analysis

- Database creation
- CSV import
- Data retrieval
- Filtering
- GROUP BY queries
- Business analysis
- Percentage calculations

5. Python EDA

- Imported libraries
- Loaded dataset
- Checked missing values
- Removed duplicates
- Converted timestamps
- Created Hour, Day and Time Slot

- Generated visualizations

6. Key Findings

- Most ride requests were completed successfully.
- Many requests failed because no cars were available.
- The City had more requests than the Airport.
- The Airport had the highest number of No Cars Available cases.
- The City had more cancelled trips.
- Peak demand occurred during specific time slots.

7. Recommendations

- Increase driver availability at the Airport.
- Reduce cancellations through driver incentives.
- Allocate drivers based on demand.
- Forecast demand during peak hours.
- Improve customer waiting times.

8. Conclusion

The project identified the main supply-demand gaps in Uber operations. Better driver allocation and demand forecasting can improve customer satisfaction and reduce cancellations.

9. References

- Uber Request Dataset
- Microsoft Excel
- MySQL Workbench
- Python
- Pandas Documentation
- Matplotlib Documentation